

Adithyan C S S

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Computer Science Undergraduate

Education

Vellore Institute of Technology, Vellore
B.Tech in Computer Science and Engineering

2024 – Present
CGPA: 9.14

Experience

Technical Intern

RF2 Digital Inc., Chennai

May 2026 – Jul 2026

- Designed, developed, and tested a C++ DSP backend engine on the Odroid N2 embedded Linux platform for a real-time audio Control Software Toolbox
- Implemented core DSP blocks including a stereo biquad IIR DC Blocker and a Compressor/Expander module with attack, hold, release, and knee-based gain reduction
- Built a dynamic **gPipeline** processing chain and a thread-safe audio queue (pthreads, mutex locking, 25-buffer pre-buffer) for glitch-free playback
- Integrated ALSA-based audio playback at 48kHz stereo with underrun (XRUN) detection and recovery
- Developed the backend side of the AlphaNexus IPC framework, including command dispatch and ACK-based responses for Start/Stop/Pause/Resume

Projects

RiderShield AI [\[GitHub\]](#)

Real-Time Collision Detection System

- Built on-device collision detection using MobileNetV2 (TFLite) on Raspberry Pi
- Achieved 15 FPS real-time inference under strict latency constraints (<100ms)
- Designed IoT pipeline: ESP32 → MQTT → FastAPI → Flutter dashboard
- Optimized model using INT8 quantization for edge deployment efficiency

AgriPrice AI [\[GitHub\]](#)

Crop Price Forecasting Platform

- Developed ensemble forecasting system using Chronos, Prophet, and regression models
- Improved prediction robustness for non-stationary agricultural price data
- Built full-stack platform (Flutter + NestJS + Python backend)
- Designed uncertainty-aware prediction with confidence intervals

RoboGuard [\[GitHub\]](#)

Predictive Fault Detection System

- Built LSTM-based model for early motor fault prediction from time-series data
- Detected anomalies 10–30 steps before failure events
- Integrated YOLOv8 for zone-based safety monitoring
- Developed real-time system using FastAPI + WebSockets

Helix [\[GitHub\]](#)

Autonomous Self-Healing Software System

- Designed and implemented the Immune System organ (scan, heal, patch, promote) including a full red-team attack pipeline against a vulnerable demo app
- Built the Immune Memory organ for vulnerability recurrence blocking using 1536-dim vector embeddings with Atlas \$vectorSearch and in-memory cosine fallback
- Integrated Qwen3.6-27B (via Groq) as the primary cognition layer for patch synthesis, drift judgement, and causal reconstruction
- Implemented a sovereign fallback routing all AI operations to a local open-weight endpoint on rate/credit exhaustion
- Wired n8n reflex arcs for automated scan-heal loops and scheduled attack runs

Technical Skills

- Languages:** Python, C, C++, Java, JavaScript, Dart
- AI/ML:** PyTorch, TensorFlow Lite, scikit-learn, YOLOv8, OpenCV
- Backend:** FastAPI, Node.js, NestJS, Redis, MongoDB, PostgreSQL, MQTT, WebSockets
- Frontend/Mobile:** Flutter, React, Tailwind CSS
- Embedded/DSP:** ALSA Audio Programming, Biquad IIR Filters, Dynamics Processing, Multithreading (pthreads), IPC Protocol Design

- **Tools:** Docker, Kubernetes, Linux, Git, Raspberry Pi, ESP32, Odroid N2

Positions of Responsibility

Chairperson *ISGF VIT Chapter* 2025 – Present

- Led student initiatives, industry collaboration, and research activities
- Organized tech quizzes, workshops, and outreach programs for 500+ students
- Mentored juniors on open-source contribution and competitive programming

Organizing Partner *Industrial Academia Conclave 2026*

- Coordinated faculty-industry discussions on applied research
- Facilitated networking sessions between 30+ industry experts and faculty members
- Managed event logistics, scheduling, and documentation for the conclave

Core Member *Backend Team, CSED VIT* 2025 – Present

- Built backend systems for large-scale student tech events
- Developed RESTful APIs and real-time features serving 1000+ concurrent users
- Collaborated with frontend and DevOps teams for seamless deployment pipelines